

SMART INDIA HACKATHON 2025

नज़र – RetailSense AI

Edge-AI Retail Intelligence Platform: Shopper Analytics, Shelf Monitoring & Queue Prediction on In-Store Cameras

Problem Statement ID [Enter PS ID]

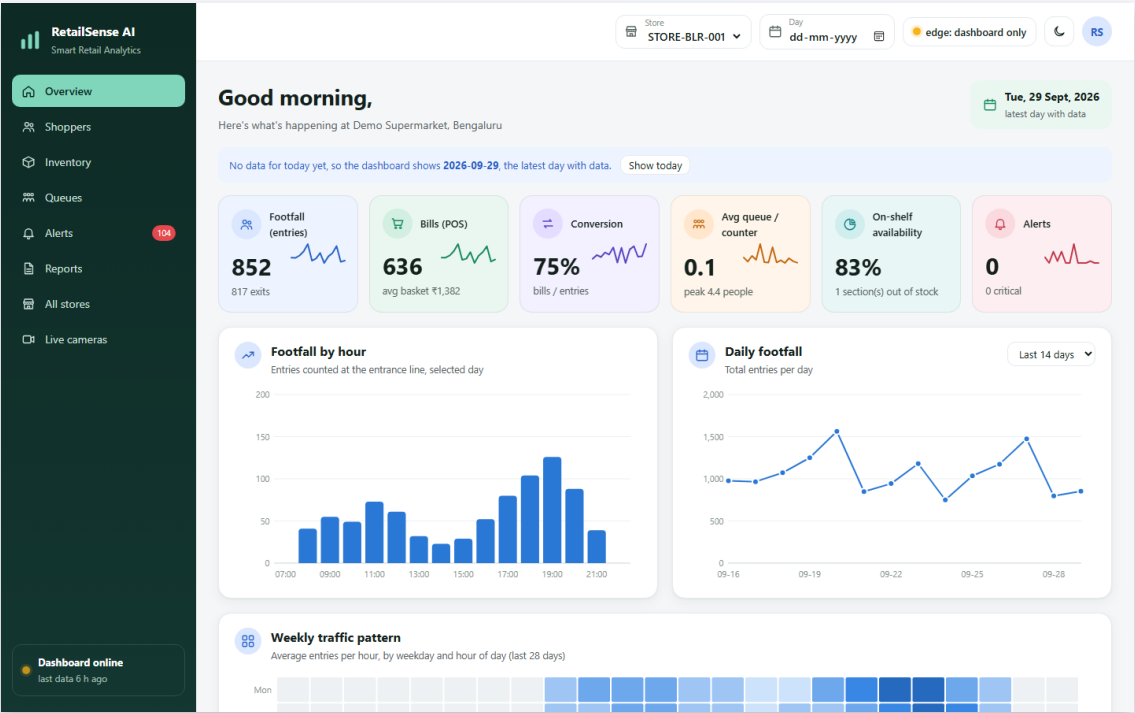
Problem Statement Title AI-powered retail intelligence with on-device (edge) AI

Theme [Enter Theme]

PS Category Software

Team ID [Enter Team ID]

Team Name [Enter Team Name]



Working prototype: live store dashboard



Understanding the Problem

- **Empty shelves go unnoticed:** about **8% of items are out of stock** at any time and shoppers **buy elsewhere**.
- **Long billing queues:** managers see the rush **too late**; shoppers **leave the line**.
- **No shopper insights:** stores **don't know footfall, peak hours** or which displays work.
- **Cloud CCTV analytics is costly:** needs **fast internet** and **sends video outside** (privacy risk).
- **Weak internet in Tier-2/3 towns:** cloud tools **stop working during outages**.



Our Solution

- **Footfall Counting:** counts people **entering and leaving** with a virtual line.
- **Heatmaps & Dwell Time:** shows **where shoppers walk and how long they stay**.
- **Shelf Monitoring:** finds **empty slots and low stock** from shelf cameras.
- **Queue Intelligence:** measures **waiting time** and **predicts queues 15 min ahead**.
- **Smart Alerts:** tells staff to **refill a shelf** or **open a counter**.
- **Edge AI:** all AI runs **inside the store**, works **without internet**.
- **Dashboard & Reports:** daily/weekly **footfall, conversion, availability**.



Our Unique Solutions

- **Works on existing CCTV:** uses the **store's own RTSP cameras**.
- **Runs on a ₹35k mini PC:** **no GPU**, 3 cameras on one CPU (measured).
- **Privacy by design:** **no faces or photos stored**, only anonymous counts.
- **Predicts, not just reports:** queue forecast from **entrance footfall**.
- **"Mark as Full" calibration:** compares a shelf with **its own full photo**.
- **Offline-first:** stores data locally, **syncs to HQ when online**.
- **POS / ERP ready:** bills via CSV/API, **auto refill orders**.

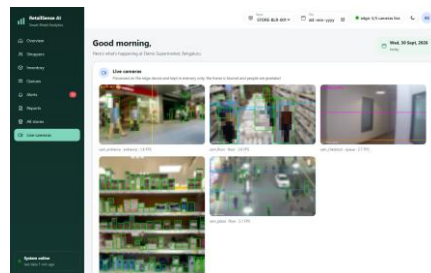
USPs

Real-time footfall,
dwell & heatmap

Empty-shelf alerts in
minutes

15-min queue
forecast + counter
advice

100% on-device,
privacy-first



Prototype (working today)

- **5 live cameras** processed on a laptop CPU
- Shelf model **mAP50 0.93** on unseen images
- Dashboard, alerts, reports, **33 tests pass**



Methodology

- Capture:** read CCTV streams (**RTSP**) at **3–10 FPS** on the in-store edge box.
- Detect:** **YOLO11** finds **people** and **shelf products** (trained on **SKU-110K**, 11,743 shelf photos).
- Track:** **ByteTrack** gives each shopper an **anonymous ID** (no face data).
- Analyse:** line counting, **zone dwell**, **heatmap**, queue length, **empty-slot** detection.
- Predict:** **LightGBM** forecasts the queue **15 min ahead**; **Erlang-C** suggests counters.
- Act:** alerts + dashboard; data kept in **SQLite (offline)** and synced to HQ.



Tech Stack



Python



PyTorch



YOLO11



OpenCV



ONNX



LightGBM / sklearn



FastAPI



SQLite



Docker



JS Dashboard



Colab T4 GPU

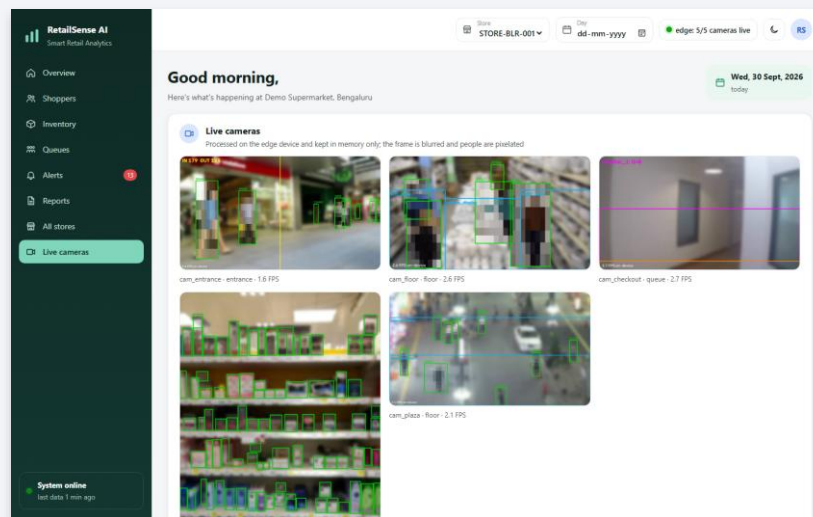


Pandas

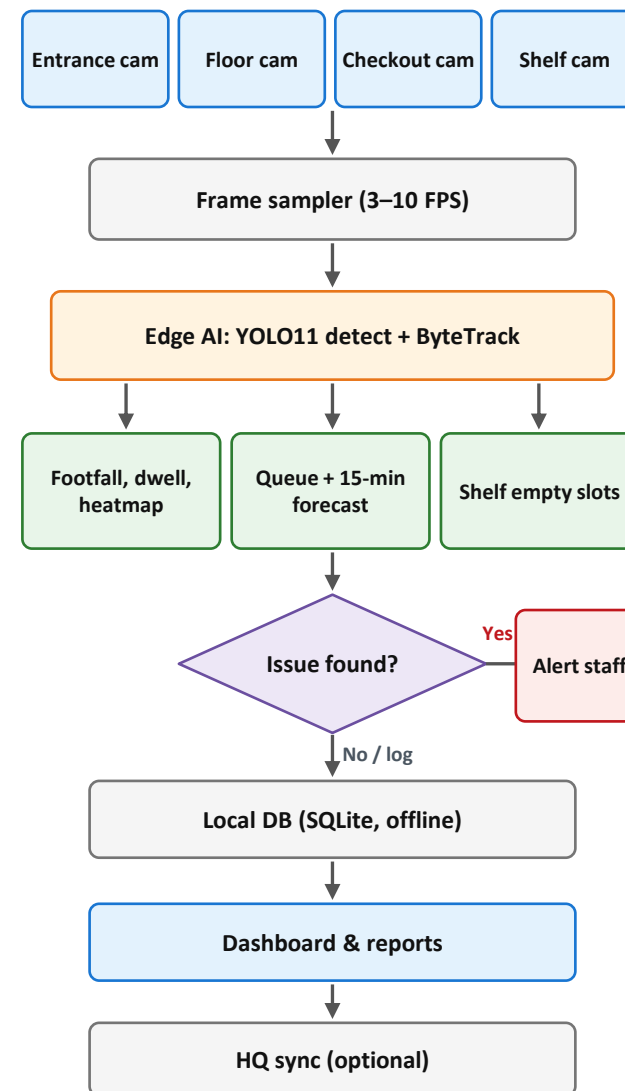
User Interface (store dashboard)



Live Cameras (edge AI output, privacy-blurred)



Implementation Flow



Prototype Output & Results



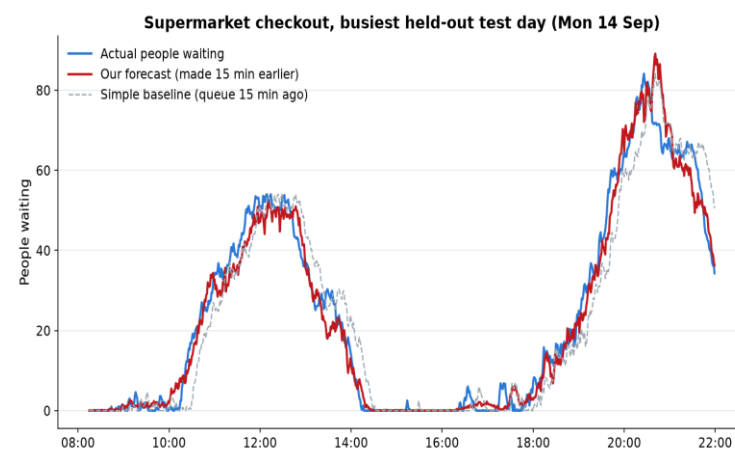
1. Footfall counting: live IN / OUT, people pixelated (MOT17 mall video)



2. Shelf monitoring: products (green), empty slots (red)



3. Shopper heatmap: where people walk and stop



4. Queue forecast on a festival rush day: error 2.8 vs 4.7 people for the baseline

Key Results (unseen test data)

0.93

mAP50 of our shelf detector on 2,936 unseen shelf photos

89%

product count accuracy per shelf photo

83%

empty-slot alerts at the right place (fixed camera)

27%

lower queue forecast error than the simple baseline

64%

footfall count accuracy on crowded street videos with human ground truth

3 cams

run together on one laptop CPU, no GPU

Impact and Benefits



Impact on Store Operations

- **Fewer empty shelves:** staff get refill alerts **before shoppers notice**.
- **Shorter queues:** open a counter **15 minutes before the rush**.
- **Better staffing:** plan shifts from **real footfall and peak hours**.
- **Smarter layout:** heatmaps show **which aisles and promos work**.
- **No manual counting:** **daily and weekly reports** are automatic.



Benefits for Shoppers & Chains

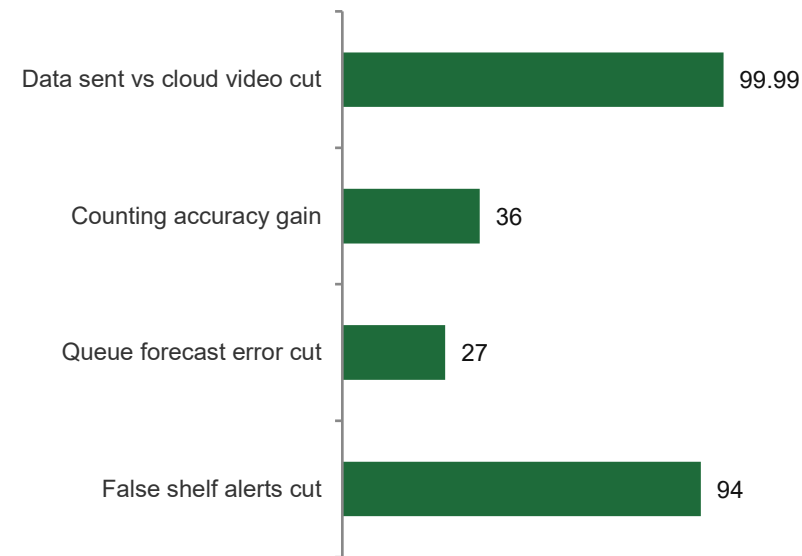
- **Products available:** shoppers find what they came for.
- **Less waiting:** faster billing, **better experience**.
- **Privacy respected:** **no faces, no photos**, only numbers.
- **Works offline:** reliable in **Tier-2/3 towns** with weak internet.
- **Multi-store view:** HQ sees **all stores on one screen**.



Economic and Strategic Gains

- **Recover lost sales:** stock-outs cost retailers **about 4% of sales** (global GMA study); faster refills win this back.
- **Low cost:** runs on **existing CCTV + a ₹35–50k mini PC**; no cloud video bills.
- **Huge data saving:** **1.2 MB/day per camera** instead of **12.6 GB** of video (**99.99% less**).
- **Big market:** India has **~13 million kirana stores** and a **US\$ 1 trillion+** retail market.
- **Future ready:** fits **DPDP Act 2023** (privacy) and scales from one shop to a **national chain**.

Improvements measured in our prototype (%)



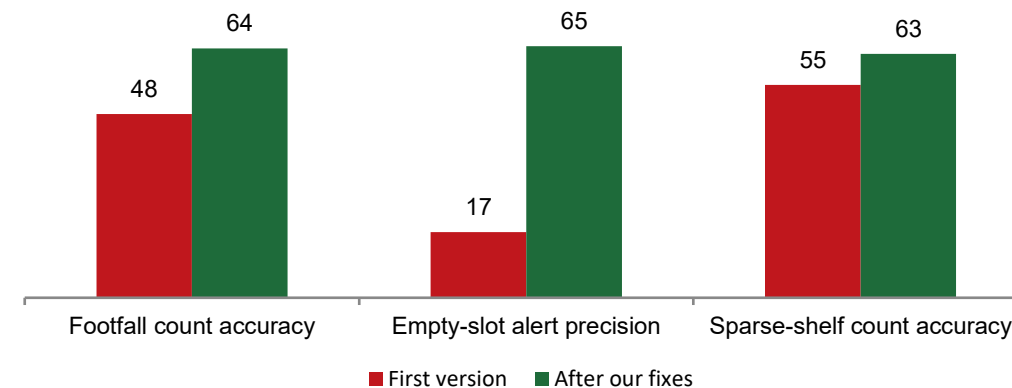
Feasibility and Viability



Analysis of the Feasibility of the Idea

- **Proven AI:** YOLO + ByteTrack are **industry standard**; our models are **trained and tested**.
- **Hardware ready:** runs on a **CPU mini PC**; Jetson / OpenVINO for more cameras.
- **Uses existing cameras:** **RTSP** from Hikvision / CP Plus / Dahua NVRs.
- **Open datasets:** **SKU-110K, COCO, MOT17**: no private data needed to start.
- **Reliable offline:** local database, **auto camera reconnect**, sync later.
- **Easy install:** **one Docker command** + zone-drawing helper tool.

Measured: before vs after our improvements (%)



WHAT IFs....?

01

Internet goes down?

Everything runs offline in the store; data syncs to HQ later.

02

A camera fails?

Other cameras keep working; the camera reconnects automatically.

03

AI makes a mistake?

Confidence limits, alert cooldown, staff can acknowledge; recalibrate with 'Mark as Full'.

04

Shoppers worry about privacy?

No faces or photos saved; anonymous IDs reset daily; clear signage.

05

Store has a small budget?

Existing CCTV + ₹35k PC; one box serves three cameras.



Strategies to Overcome Challenges

- **Low computing power:** light **YOLO11n**, **3–10 FPS** sampling, higher resolution only for the entrance camera.
- **Limited Indian retail data:** start with **open datasets**, then **fine-tune on store photos** using free **Colab GPU**.
- **Crowded entrances:** angled camera over the door, **960 px** input, **jitter-proof counting** (tested on MOT17).
- **Counting bias on odd shelves:** per-camera **'Mark as Full' baseline** cancels the bias.

Research & References

Feature	Our Proposed Solution	Existing Conventional Solutions
Where AI runs	Edge: inside the store, on a small PC	Cloud servers far away
Internet need	Works offline; syncs numbers later	Needs constant, fast internet
Privacy	No faces / photos stored; anonymous counts	Video uploaded to the cloud
Cost	Existing CCTV + ₹35–50k PC	Subscription + high bandwidth bills
Shelf checking	Automatic empty-slot alerts	Manual audits / staff rounds
Billing queues	Predicts 15 min ahead, advises counters	Reacts after the queue forms
Shopper insights	Footfall, dwell, heatmap, conversion	Only CCTV recording



Research and Dataset References

- **SKU-110K** — dense retail shelf dataset (CVPR 2019)
https://github.com/eg4000/SKU110K_CVPR19
- **MOT17** — people-tracking benchmark with ground truth
<https://motchallenge.net/data/MOT17/>
- **COCO** — person detection dataset
<https://cocodataset.org>
- **ByteTrack** — multi-object tracking (ECCV 2022)
<https://arxiv.org/abs/2110.06864>
- **Ultralytics YOLO11** — real-time detector
<https://docs.ultralytics.com>
- **Out-of-Stock Detection** — Sensors 2024 (shelf voids)
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10819825/>
- **Retail Out-of-Stocks** — worldwide study, GMA 2002
https://www.supplychain247.com/images/pdfs/GMA_2002_Worldwide_OOS_Study.pdf
- **Kirana modernisation** — Invest India
<https://www.investindia.gov.in/team-india-blogs/modernization-kirana-stores-india>
- **DPDP Act 2023** — India data-protection law
<https://www.meity.gov.in/data-protection-framework>
- **IBEF Retail India** — Indian retail market data
<https://www.ibef.org/industry/retail-india>
- **Erlang-C queueing model** — counters needed for a target wait
[https://en.wikipedia.org/wiki/Erlang_\(unit\)](https://en.wikipedia.org/wiki/Erlang_(unit))